File handling in C Programming

What is the advantage of File handling?

The operations used in a C program are done on a terminal which is not stored anywhere. The output is deleted as soon as the program is closed. But in the software industry, most programs are written to store the information fetched from the program. File handling is beneficial in such cases as information can be stored in files.

The data stored in the file can be accessed, updated, and deleted anywhere and anytime providing high reusability.

Some file handling functions used in C Programming are:

fopen( ) function in C is a library function that is used to open a file to perform various operations which include reading, writing, etc. along with various modes.

fclose( ) is a library function used to close a file that was previously opened using fopen( ).

fscanf( ) is a library function that uses formatted string and variable arguments list to take input from a file.

fgets( ) is a library function that is used to input the whole line from the file.

fgetc( ) is a library function that is used to read a single character from the file.

fread( ) is a library function that is used to read the specified bytes of data from a binary file.

fprintf( ) is a library function that uses formatted string and variable arguments list to print output to the file.

fputs( ) is a library function that is used to print the whole line in the file and a newline at the end.

fputc( ) is a library function that is used to print a single character into the file.

fwrite( ) is a library function that is used to write the specified amount of bytes to the binary file.

fseek( ) is a library function that is used to move the file pointer associated with a given file to a specific position.

feof( ) is a library function that is used to check whether the file pointer to a stream is pointing to the end of the file or not. It returns a non-zero value if the end is reached, otherwise, it returns 0.

Different File Opening ~~Codes~~ Modes :-

1) "r" — Searches file. If the file is opened successfully, fopen() loads it into memory & sets up a pointer which points to the first character in it. If the file cannot be opened, fopen() returns NULL.

Operations possible are reading from the file.

2) "w" — Searches file. If the file exists, its contents are overwritten. If the file does not exist, a new file is created. If the file cannot be opened, fopen() returns NULL.

Operations possible are writing to the file.

3) "a" — Searches file. If the file is opened successfully, fopen() loads it into memory and sets up a pointer that points to the last character in it. If the file does not exist, a new file is created. If the file cannot be opened, fopen() returns NULL.

Operations possible are adding new contents at the end of file.

4) "r+" — Searches file. If opened successfully, fopen() loads it into memory and sets up a pointer which points to the first character in it. If the file cannot be opened, fopen() returns NULL.

Possible operations are reading existing contents, writing new contents, modifying existing contents of the file.

5) "w+" — Searches file. If the file exists, its contents are overwritten. If the file does not exist, a new file is created. If file cannot be opened then NULL is returned.

Scanned by CamScanner

Possible operations are writing new contents, reading them back and modifying existing contents of the file.

6) "a+" — Searches file. If the file is opened successfully, fopen() loads it into memory and sets up a pointer which points to the first character in it. If the file does not exist, a new file is created. If the file cannot be opened, then NULL is returned.

Possible operations are reading existing contents, appending new contents to end of file. Cannot modify existing contents.

Scanned by CamScanner

Some programs on File Handling done in class

Write a C program to read the contents of a text file and display the contents in the output screen.

```
#include<stdio.h>
#include<stdlib.h>

void main( )
{
    FILE *fp;
    char ch;

    fp=fopen("source.txt","r");

    if(fp==NULL)
    {
            printf("\nCannot open file.");
            exit(0);
    }

    while(1)
    {
             ch=fgetc(fp);
             if(ch==EOF)
                break;
             printf("%c",ch);
    }

    fclose(fp);
}
```

Write a C program to take a string as input from the user and write it in a text file.

```
#include<stdio.h>
#include<stdlib.h>

void main( )
{
    FILE *fp;
    char str[20];

    fp=fopen("source.txt","w");
```

```
    if(fp==NULL)
    {
            printf("\nCannot open file.");
            exit(0);
    }

    printf("Enter the string\n");
    scanf("%[^\n]s",str);

    fputs(str,fp);

    fclose(fp);
}
```

Write a C program to read the contents of a text file and write those contents in another text file (or Write a C program to copy the contents of a text file in another text file).

```
#include<stdio.h>
#include<stdlib.h>

void main( )
{
    FILE *ft,*fs;
    char ch;

    fs=fopen("source.txt","r");

    if(fs==NULL)
    {
            puts("\nCannot open file.");
            exit(0);
    }

    ft=fopen("target.txt","w");

    if(ft==NULL)
    {
            puts("\nCannot open file.");
            fclose(fs);
            exit(0);
    }
```

```
    while(1)
    {
            ch=fgetc(fs);

            if(ch==EOF)
            {
               break;
            }
            else
            {
              fputc(ch,ft);
            }
    }

    fclose(fs);
    fclose(ft);
}
```